



TACIT TEACHER

PUTTING SCIENTISTS DIRECTLY IN THE DRIVER'S SEAT

[TEAM]

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[MISSION]

Enable scientists to onboard robotic workflows easily



SYSTEM BOTTLENECK

The Wicked Hard Problem

Adapting complex human protocols to robots

Lost Expertise: Tacit knowledge is difficult to extract and transfer to robots:

- force feedback (touch/feel)
- adaptive actions (intuition)
- smell/sound???

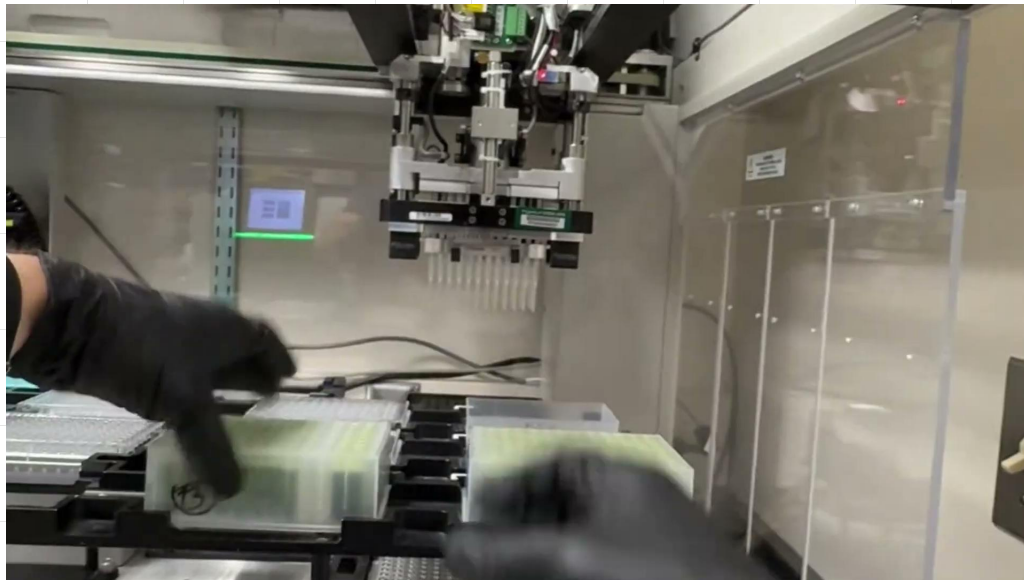


Plate Peeling

CURRENT GAPS

- Many workflows **need rubber mat seals** that can't be avoided
- **Teaching scientists** how to peel successfully **is not easy** (try it!)
- **Peeling is often step 1** and unlocking it can enable autonomous experiments
- A mechanical peeling robotic skill **does not exist**

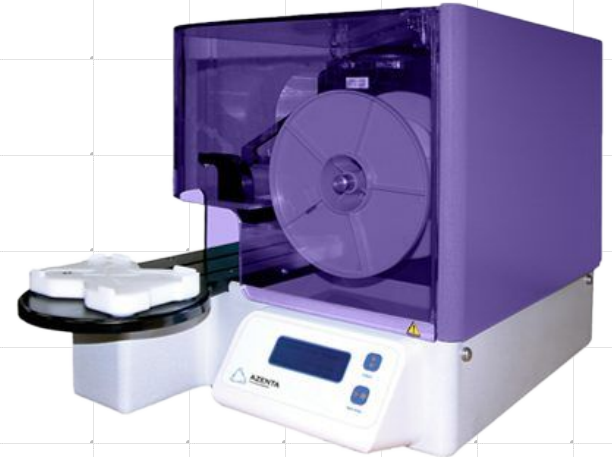
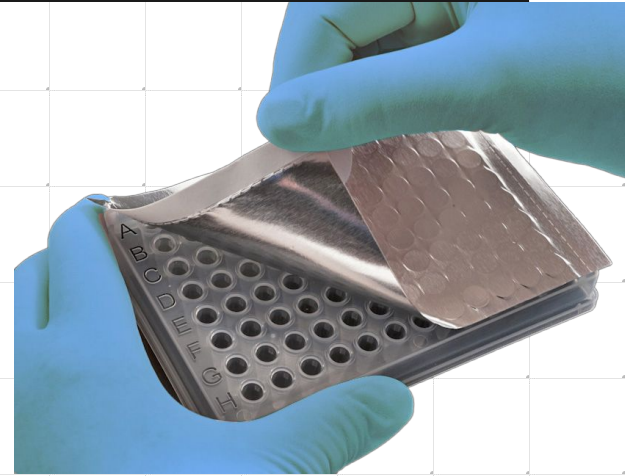




A Problem 'Too Hard' for Traditional Automation

Only one lab automation device exists (XPeel):

- only handles aluminum stickers
- specific low-profile labware
- fails often
- needs babysitting





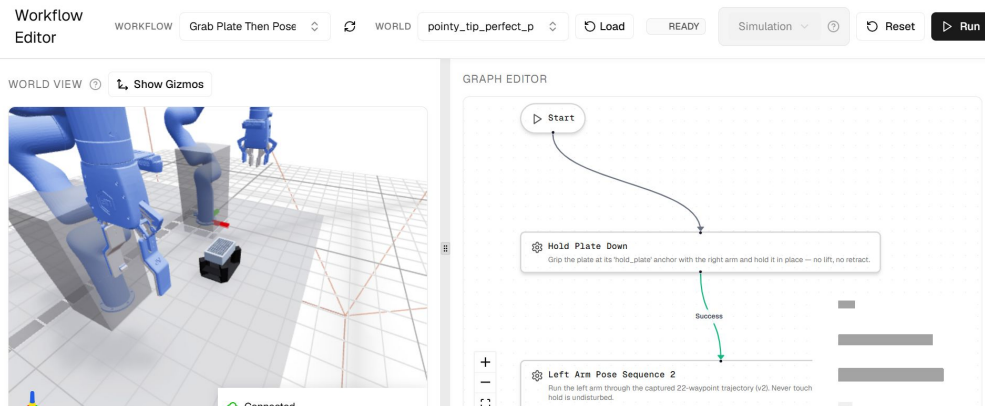
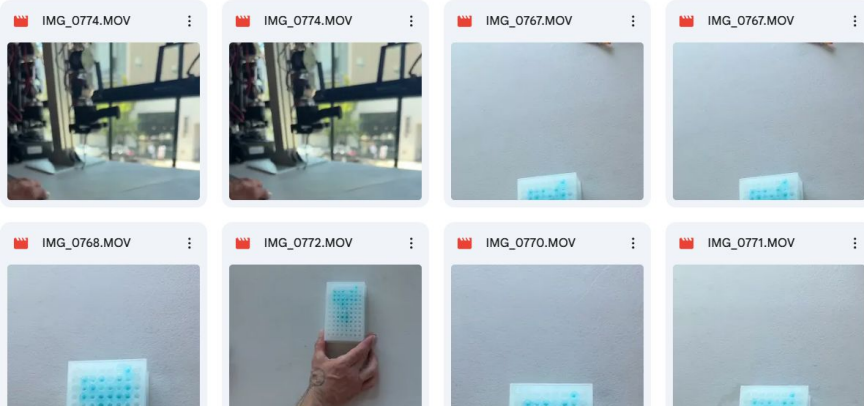
Our Approach

Videos
with
Narration
success, failures

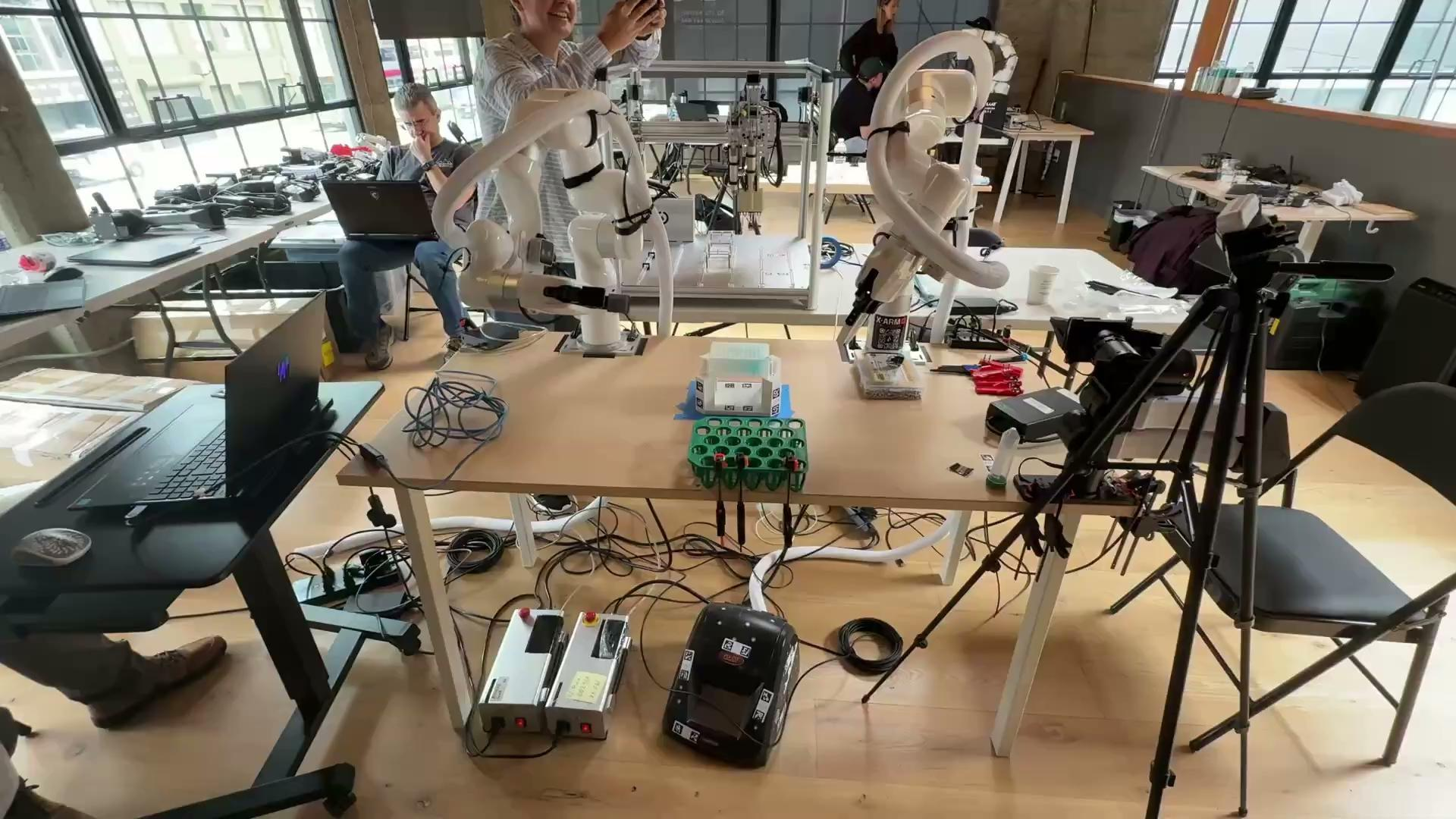
Agentic
Interview

Zeon
Workflow

Teach &
Test



Introducing ZPeel!

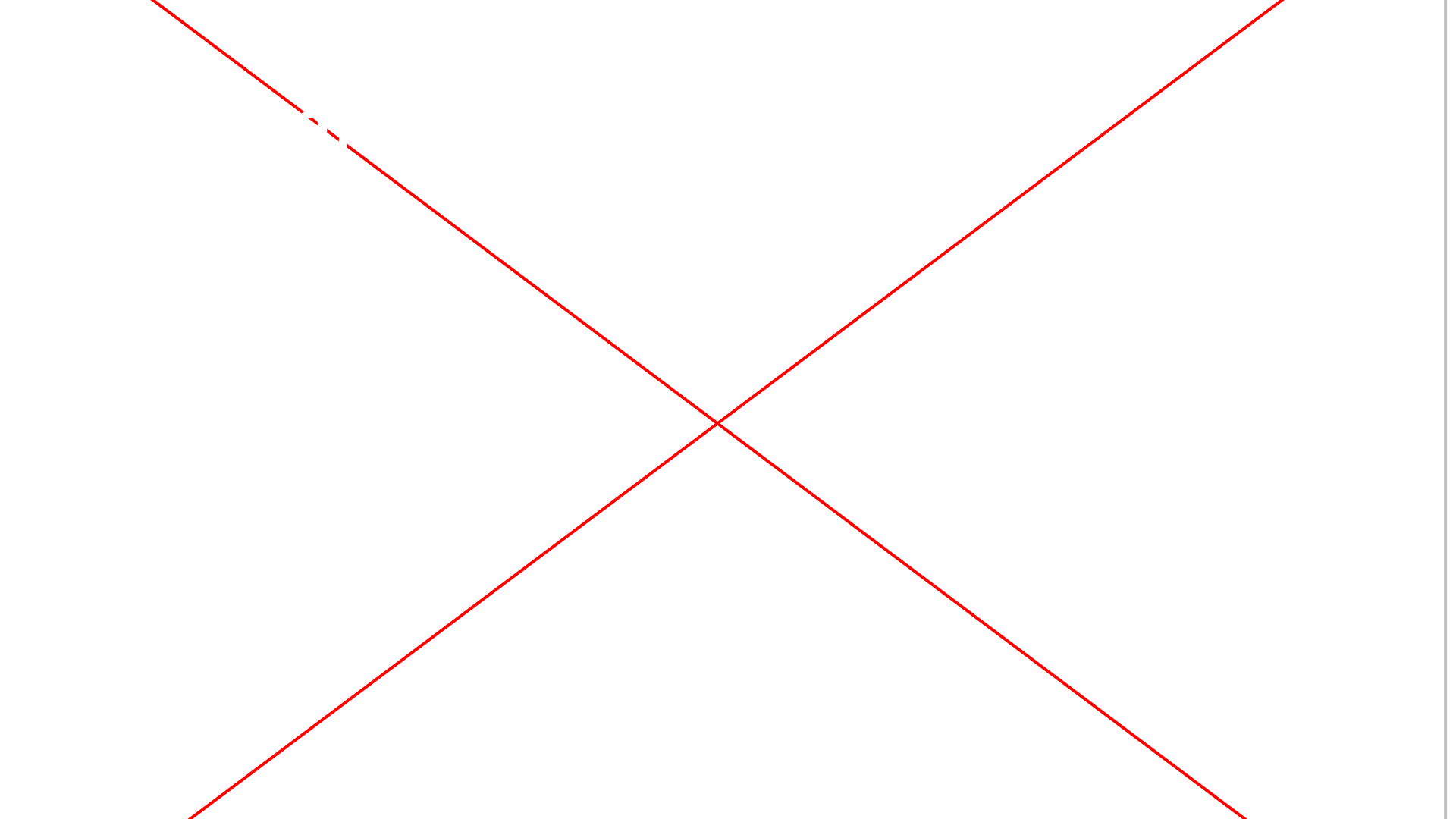


Jobs for Tacit Teacher

- Tissue grinding
- Loading centrifuge
- Protein extraction
- Vortexing
- Transfer labware between stations

Video at 32X 🤖







So What?

[01] MASTERY

a hard dexterous lab task

[02] EXTENSIBILITY

Can adapt to other tasks like opening tubes, manipulating powders, etc

[03] OPERATIONAL AUTONOMY

Scientists can create custom skills without external help

[04] INVESTMENT EFFICIENCY

Eliminates expensive, uni-tasking hardware

